12/04/2024

Phase-I
CODE-A

Aakash

Medical | IIT-JEE | Foundations

Corporate Office: Aakash Tower, 8, Pusa Road, New Delhi-110005, Ph.011-47623456

FINAL TEST SERIES for NEET-2024

MM : 720

Test - 7

Time : 3 Hrs. 20 Mins.

Answers

1. (4)	41. (3)	81. (3)	121. (4)	161. (3)
2. (4)	42. (2)	82. (4)	122. (2)	162. (4)
3. (1)	43. (1)	83. (1)	123. (1)	163. (2)
4. (1)	44. (2)	84. (2)	124. (4)	164. (3)
5. (3)	45. (2)	85. (4)	125. (2)	165. (4)
6. (2)	46. (1)	86. (4)	126. (3)	166. (2)
7. (1)	47. (2)	87. (1)	127. (2)	167. (3)
8. (1)	48. (1)	88. (3)	128. (1)	168. (1)
9. (4)	49. (1)	89. (1)	129. (3)	169. (1)
10. (1)	50. (1)	90. (2)	130. (3)	170. (4)
11. (4)	51. (3)	91. (4)	131. (3)	171. (2)
12. (2)	52. (1)	92. (3)	132. (2)	172. (2)
13. (1)	53. (1)	93. (2)	133. (3)	173. (3)
14. (1)	54. (3)	94. (2)	134. (1)	174. (4)
15. (2)	55. (3)	95. (4)	135. (4)	175. (4)
16. (2)	56. (1)	96. (1)	136. (2)	176. (2)
17. (1)	57. (2)	97. (2)	137. (1)	177. (3)
18. (3)	58. (1)	98. (4)	138. (1)	178. (3)
19. (4)	59. (1)	99. (3)	139. (2)	179. (3)
20. (4)	60. (2)	100. (2)	140. (3)	180. (3)
21. (1)	61. (4)	101. (4)	141. (1)	181. (2)
22. (1)	62. (3)	102. (2)	142. (2)	182. (4)
23. (1)	63. (3)	103. (2)	143. (1)	183. (2)
24. (4)	64. (1)	104. (4)	144. (3)	184. (2)
25. (3)	65. (4)	105. (1)	145. (4)	185. (4)
26. (1)	66. (1)	106. (2)	146. (2)	186. (3)
27. (3)	67. (2)	107. (4)	147. (3)	187. (3)
28. (4)	68. (2)	108. (3)	148. (2)	188. (2)
29. (1)	69. (3)	109. (1)	149. (4)	189. (3)
30. (4)	70. (2)	110. (3)	150. (2)	190. (1)
31. (2)	71. (3)	111. (2)	151. (3)	191. (4)
32. (1)	72. (2)	112. (1)	152. (4)	192. (2)
33. (3)	73. (2)	113. (3)	153. (3)	193. (3)
34. (4)	74. (2)	114. (1)	154. (1)	194. (2)
35. (1)	75. (2)	115. (2)	155. (2)	195. (3)
36. (3)	76. (4)	116. (4)	156. (3)	196. (3)
37. (3)	77. (2)	117. (1)	157. (2)	197. (3)
38. (3)	78. (2)	118. (3)	158. (3)	198. (2)
39. (3)	79. (1)	119. (4)	159. (3)	199. (3)
40. (3)	80. (4)	120. (3)	160. (1)	200. (3)

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Answers and Solutions

PHYSICS

SECTION - A

1. Answer (4)

   Power factor, $\cos\phi = \frac{R}{Z}$. It has no unit

2. Answer (4)

   Direction of propagation of wave is parallel to $\vec{E} \times \vec{B}$

3. Answer (1)

   For an LR d.c. circuit

   $R = \frac{V}{I} = \frac{100}{1} = 100\ \Omega$

   For an RL. AC. Circuit

   $\sqrt{\omega^2 L^2 + R^2} = \frac{E}{I} = \frac{100}{0.5} = 200\ \Omega$

   $(100\ \pi)^2 L^2 + (100)^2 = (200)^2$

   $(10^5)\pi L^2 = 200^2 - 100^2$

   $L^2 = \frac{(300)(100)}{(10^4)\pi}$

   $L = \frac{\sqrt{3}}{\pi}\text{H}$

4. Answer (1)

   We know, at resonance $\omega_0 = \frac{1}{\sqrt{LC}}$

   $\therefore \sqrt{L_1C_1} = \sqrt{L_2C_2}$

   $\sqrt{L_1C_1} = \sqrt{L_2\frac{C_1}{4}}$

5. Answer (3)

   It is correct that laminated core is used to reduce eddy currents which in turn reduces energy loss due to it (Iron loss).

6. Answer (2)

   We know, displacement current, $I_d = \varepsilon_0 \frac{d\phi_E}{dt}$

   or $I_d = \varepsilon_0 A\left(\frac{dE}{dt}\right)$

   $I_d \propto \frac{dE}{dt}$

   $I_d \propto \frac{d}{dt}(4t^2)$

   $I_d \propto 8t$ or $I_d \propto t$

7. Answer (1)

   We know $\frac{E_0}{B_0} = c \Rightarrow \frac{30}{B_0} = 3\times10^8$

   $B_0 = 10^{-7}\ \text{T}$

8. Answer (1)

   Ultraviolet rays are used to kill germs thereby disinfecting the water.

9. Answer (4)

   As light is moving from rarer to denser medium, thus it will never undergo total internal reflection.

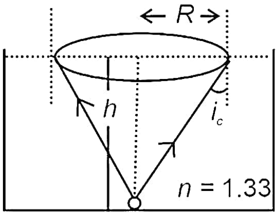
10. Answer (1)

We know, Angular magnification , $M = \frac{f_0}{f_e} = \frac{15}{0.01}$

$M = 1500$

11. Answer (4)

$A = \pi R^2 = \frac{\pi h^2}{n^2 - 1}$



$= \frac{3.14}{\frac{16}{9} - 1} = \frac{(3.14)9}{7} \approx 4\text{m}^2$

12. Answer (2)

We know, $\sin\theta = \frac{1}{y_{n_x}} = \frac{n_y}{n_x} = \frac{v_x}{v_y}$

$v_y = \frac{v_x}{\sin\theta} \text{or} \frac{v}{\sin\theta}$

13. Answer (1)

Here,

Shift, $x = t\left(1 - \frac{1}{\mu}\right)$

$0.5 = t\left(1 - \frac{1}{2}\right)$

$0.5 = \frac{t}{2}$

$t = 1$ cm

14. Answer (1)

Due to introduction of plate the optical path will become μt

Thus increase = $\mu t - t = (\mu - 1)t$

15. Answer (2)

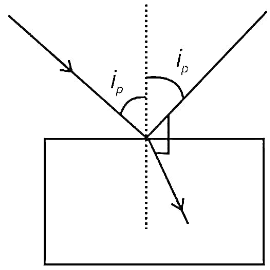
When $a_1 = a_2 = a$(say)

In YDSE

Then for maxima $I \propto (2a)^2 = 4a^2$ and for minima $I \propto 0$

Thus there is a good contrast between maxima and minima but $I \propto a^2$ is not the reason for this

16. Answer (2)



By Brewster's law $\mu = \tan i_p$

In such case, reflected ray will be completely polarised while refracted (transmitted) one will be partially polarised.

17. Answer (1)

$Z = \sqrt{(X_L - X_C)^2 + R^2}$

$\because X_L - X_C = 0$

$\therefore Z = R$

$i = \frac{V}{R} = \frac{110}{55}$

$i = 2$ A

18. Answer (3)

Angular width = $2\theta = \frac{2\lambda}{a}$

$= \frac{2 \times 500 \times 10^{-10}}{10 \times 10^{-5} \times 10^{-2}} = \frac{10 \times 10^{-7}}{10 \times 10^{-7}} = 1\,\text{rad}$

19. Answer (4)

For AC, the average value of current for complete cycle will be zero.

20. Answer (4)

β-rays do not travel with speed of light, as they are not electromagnetic waves.

21. Answer (1)

Out of given options, Radio wave has largest wavelength

Range : 10^{-2} m to 10^4 m.

22. Answer (1)

$F = \frac{IA}{c} = \frac{30 \times 2}{3 \times 10^8} = 2 \times 10^7$ N

$\Delta P = F\Delta t = 2 \times 10^{-7} \times 10 = 2 \times 10^{-6}$

23. Answer (1)

$z = \sqrt{x_L^2 + R^2} = \sqrt{6^2 + 6^2}$

$z = 6\sqrt{2}\ \Omega$

24. Answer (4)

rms value $= \sqrt{\int_0^3 \frac{9t^2 dt}{3}} = 5.19$ A

25. Answer (3)

In step up transformers, Number of turns in secondary is more than that in primary and this transformer increases the voltage but decreases the current. In step down transformers, Number of turns in secondary is less than that in primary and this transformer decreases the voltage but increases the current.

26. Answer (1)

Conversion factor of transformer = $N_s/N_p = \frac{100}{20} = 5$

Thus, $I_s = \frac{I_p}{5} = \frac{20}{5} = 4$ A

27. Answer (3)

$$X_C = \frac{1}{\omega C} = \frac{1}{2\pi f C}$$

$\Rightarrow \quad X_C \propto \frac{1}{f} \rightarrow$ Rectangular hyperbola

28. Answer (4)

As power factor for an ideal inductor is zero. Thus power consumed is zero.

29. Answer (1)

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{v} + \frac{1}{-60} = \frac{1}{-20}$$

$$\therefore \quad \frac{1}{v} = -\frac{1}{20} + \frac{1}{60} = \frac{-3+1}{60}$$

$v = -30$ cm

30. Answer (4)

When object is placed as centre of curvature then its image is formed at centre of curvature *i.e.*, distance is zero.

31. Answer (2)

$m_T = m_1 \times m_2$

$96 = 8 \times m_2$

$m_2 = 12$

32. Answer (1)

$\sin\theta \approx \theta$

$a \sin\theta = (2n + 1)\frac{\lambda}{2} = (2 \times 1 + 1)\frac{\lambda}{2} = \frac{3\lambda}{2}$

$$a\theta = \frac{3\lambda}{2}$$

$$\theta = \frac{3\lambda}{2a} = \frac{3 \times 600 \times 10^{-9}}{2 \times 0.25 \times 10^{-3}} = 3.6 \times 10^{-3} \text{ rad}$$

33. Answer (3)

Diffraction and polarization are both wave phenomenon.

34. Answer (4)

$I = I_0 \cos^2\theta$

$$\cos^2\theta = \frac{I}{I_0} = \frac{1}{2}$$

$\cos\theta = \pm\frac{1}{\sqrt{2}}$, $\theta = 45°$ or $135°$

35. Answer (1)

First minima $\sin\theta = \frac{\lambda}{a} = \theta$

For first maxima $\sin\theta' = \frac{3\lambda'}{2a}$

As two coincide

$\theta = \theta'$

$$\frac{3\lambda_2}{2a} = \frac{\lambda_1}{a}$$

$$\therefore \quad \lambda_2 = \frac{2}{3} \times (\lambda)_1$$

$$= \frac{2}{3} \times 660 = 440 \text{ nm}$$

SECTION-B

36. Answer (3)

$$\mu = \frac{\sin\left(\frac{\delta_m + A}{2}\right)}{\sin\frac{A}{2}} = \frac{\sin A}{\sin\frac{A}{2}} = \frac{2\sin\frac{A}{2}\cos\frac{A}{2}}{\sin\frac{A}{2}}$$

$$2\cos\frac{A}{2} = \mu = \sqrt{3}$$

$$\cos\frac{A}{2} = \frac{\sqrt{3}}{2}$$

$$\therefore \quad \frac{A}{2} = 30°$$

$$A = 60°$$

37. Answer (3)

Dispersion is due to change in refractive index with wavelength

38. Answer (3)

Wave velocity decreases, frequency remains same and wavelength decreases.

39. Answer (3)

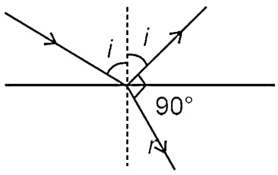
$I_R \propto (a_1^2 + a_2^2 + 2a_1a_2\cos\phi)$

$I_R \propto (3A)^2 + (2A)^2 + 2\times 3A\times 2A\cdot\cos\frac{\pi}{3}$

$\propto 9A^2 + 4A^2 + 6A^2$

$\therefore \quad I_R \propto 19A^2$

40. Answer (3)



By Brewster' Law

$\mu = \tan i$

$\frac{c}{v} = \tan i$

$\therefore \quad \tan i = \frac{c}{v} = \frac{3\times 10^8}{2.25\times 10^8}$

$\Rightarrow \quad \tan i = \frac{4}{3}$

$\therefore \quad i = 53°$

41. Answer (3)

$\mu = \dfrac{\sin\left(\frac{A+\delta}{2}\right)}{\frac{A}{2}}$

For small angle, $\mu = \dfrac{\left(\frac{A+\delta}{2}\right)}{\frac{A}{2}}$

$\mu A = A + \delta$

$\therefore \quad \delta = \mu A - A = (\mu - 1)A$

42. Answer (2)

$\sin\phi = \frac{100-60}{50}$

$\sin\phi = \frac{40}{50}$

$\sin\phi = \frac{4}{5}$

$\phi = \sin^{-1}\left(\frac{4}{5}\right)$

43. Answer (1)

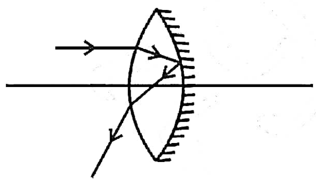
A hot wire instrument reads RMS values of both AC and DC.

44. Answer (2)

The concave lens of water has negative power, so overall power decreases.

45. Answer (2)

$\frac{1}{f_{net}} = \frac{-2}{f_L} + \frac{1}{f_m}$



$\frac{1}{f_{net}} = \frac{-2(3)}{80} + \frac{1}{-20}$

$\frac{1}{f_{net}} = \frac{-6-4}{80} \Rightarrow f_{net} = -8\,\text{cm}$

46. Answer (1)

We know,

$F = pA$

$F = \frac{2IA}{C}$ (for perfectly reflecting surface)

$F = \frac{100\times 6\times 10^{-4}\times 2}{3\times 10^8}$

$F = 4\times 10^{-10}$ N

47. Answer (2)

$z = \sqrt{R^2 + (X_L - X_C)^2}$

$z = \sqrt{100^2 + (200-100)^2} = 100\sqrt{2}\ \Omega$

$I = \frac{V}{z} = \frac{150\sqrt{2}}{100\sqrt{2}} = \frac{3}{2}$ A

$P = I^2R = \frac{9}{4}\times 100 = 225$ W

48. Answer (1)

Impedance has unit of resistance (Ω)

Wattless current is actually current (A)

Q-factor is a ratio and is thus unitless

RMS voltage is potential difference (V)

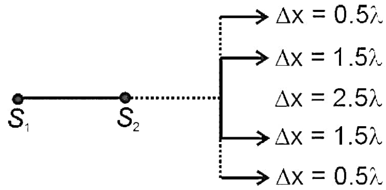
49. Answer (1)

We know, $y_5 = \dfrac{5\lambda D}{d} = 2\text{ cm}$

$y_3' = \dfrac{5\lambda D}{2d} = \dfrac{1}{2}(y_5)$

$y_3' = 1\text{ cm}$

50. Answer (1)



∴ Number of minima = 5

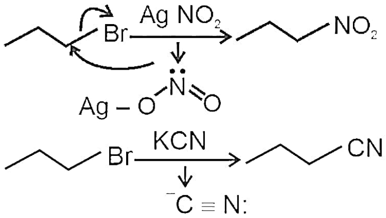
CHEMISTRY

SECTION - A

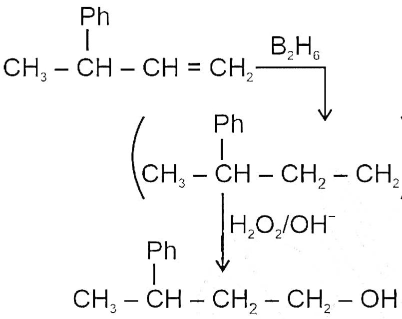
51. Answer (3)

S_N1 reactions undergo racemization.

52. Answer (1)

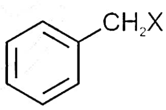


53. Answer (1)

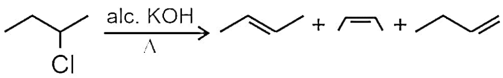


54. Answer (3)

Benzylic halide are the compounds in which the halogen atom is bonded to an sp^3 – hybridized carbon atom attached to an aromatic ring.



55. Answer (3)



56. Answer (1)

In polar aprotic medium anion remains unsolvated.

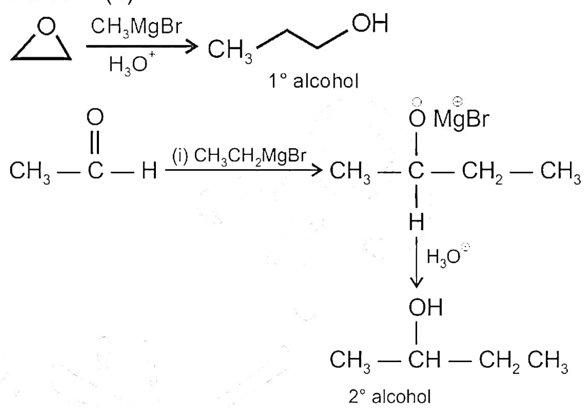
57. Answer (2)

$-CCl_3$ is m-directing group.

58. Answer (1)

Phenols reacts with aqueous NaOH to form sodium phenoxide.

59. Answer (1)



60. Answer (2)

Electron donating group at para position of phenol decreases its acidity and increases pK_a.

61. Answer (4)

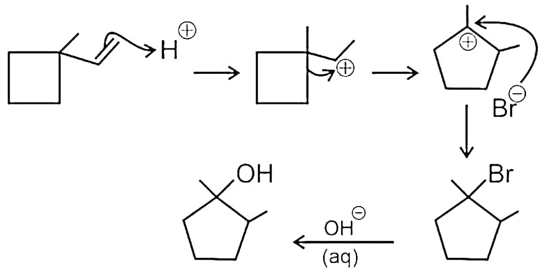
Ether oxygen which is connected to both side sp^2 hybrid carbon will not be cleaved easily by heating with HI due to double bond character of C – O bonds.

62. Answer (3)

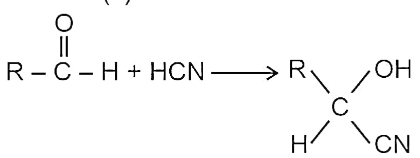
Williamson synthesis:

$$R-X+R'-O^-Na^+ \xrightarrow{S_N2} R-O-R'+NaX$$

63. Answer (3)



64. Answer (1)



Optically active compond

65. Answer (4)

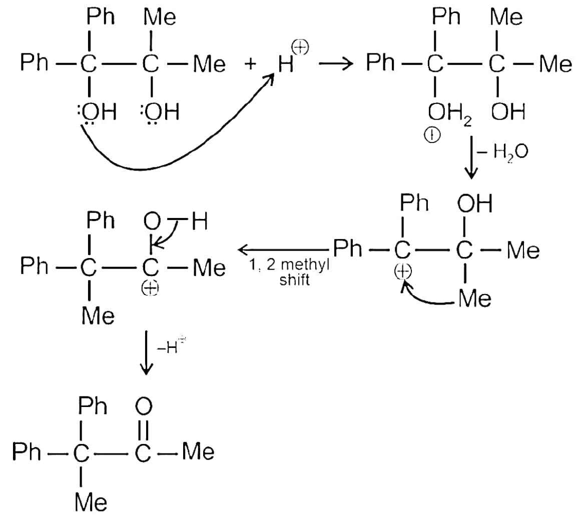
$$C_2H_5-C\equiv CH \xrightarrow{CH_3MgBr} C_2H_5-C\equiv CMgBr + CH_4$$

$$\xrightarrow{CH_3Br} C_2H_5-C\equiv C-CH_3$$

66. Answer (1)

Allylic bromination takes place by NBS/hν

67. Answer (2)



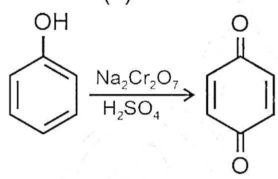
68. Answer (2)

Polar protic solvent $\rightarrow$ H_2O, Alcohol

Polar aprotic solvent $\rightarrow$ DMSO

Non-polar $\rightarrow$ CS_2

69. Answer (3)



benzoquinone

70. Answer (2)

In Fittig reaction aryl halide treated with Na in dry ether, in which two aryl groups are joined together.

71. Answer (3)

Reactivity order towards nucleophilic addition reaction :

Aldehydes > aliphatic ketones > aromatic ketones

72. Answer (2)

Aldol condensation reaction:

$$2CH_3CHO \xrightarrow{\text{dil. NaOH}} CH_3-\underset{}{\overset{OH}{\overset{|}{C}H}}-CH_2CHO$$

$$\xrightarrow[\Delta]{} \underset{\substack{\text{Crotonaldehyde}\\ \text{[A]}}}{CH_3-CH=CHCHO}$$

73. Answer (2)

$$CH_3COONa + H_2O \xrightarrow{\text{electrolysis}}$$

$$C_2H_6 + CO_2 + H_2$$

74. Answer (2)

$$\underset{\text{Acetaldehyde}}{CH_3CH=O} + HCN \longrightarrow \underset{\text{Cyanohydrin}}{CH_3-CH(OH)(CN)}$$

$$\xrightarrow{H_3O^+/\Delta} \underset{\text{Lactic acid}}{CH_3-CH(OH)(COOH)}$$

75. Answer (2)

Oxime is formed when NH_2OH react with aldehydes or ketones.

76. Answer (4)

$$R-Mg-X + CO_2 \xrightarrow[(2)\ H_3O^+]{(1)\text{ dry ether}} R-COOH$$

77. Answer (2)

On the basis of stability of conjugate base of acids, the order of acidic strength is: a > b > c

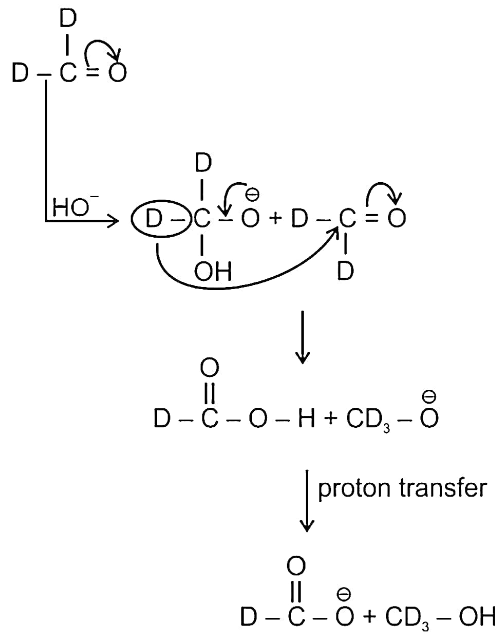
78. Answer (2)

$$R-COZ \xrightarrow{H_2O} R-COOH + HZ$$

- More the leaving ability of Z, more will be the reactivity of acid derivative towards hydrolysis
- Leaving ability:
  $X^- > RCOO^- > RO^- > NH_2^-$

79. Answer (1)

Cannizzaro reaction:



80. Answer (4)

Hell-Volhard-Zelinsky reaction:

$$CH_3CH_2COOH \xrightarrow{\text{Red P}/Br_2} \underset{\text{(Acylhalide)}}{CH_3CH_2-COBr.}$$

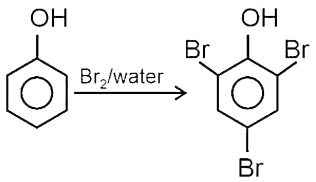
$$CH_3CH_2COBr \xrightarrow{Br_2} CH_3\underset{\substack{|\\ Br}}{CH}-COBr + HBr$$

$$CH_3\underset{\substack{|\\ Br}}{CH}-COBr \xrightarrow{H_2O} CH_3\underset{\substack{|\\ Br}}{\overset{\alpha}{CH}}-COOH$$

81. Answer (3)

Tertiary alcohol reacts with Lucas reagent (conc. HCl and $ZnCl_2$) most readily by S_N1 mechanism.

82. Answer (4)



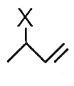
83. Answer (1)

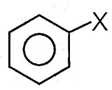
$C_6H_5\overset{\oplus}{C}H_2$ is less stable than $(CH_3)_3\overset{\oplus}{C}$ hence $(CH)_3CCl$ will give faster S_N1 reaction

84. Answer (2)


Alkyl halide


Vinyl halide


Allyl halide


Aryl halide

85. Answer (4)

Rate of S_N2 reaction follows as:

$CH_3-Cl >$  $Cl >$  Cl

SECTION-B

86. Answer (4)

Molecule (4) has plane of symmetry hence it is optically inactive.

87. Answer (1)

$$C_6H_5MgBr + CH_3-\overset{\substack{CH_3\\ |}}{\underset{\substack{|\\ OH}}{C}}-CH_3 \longrightarrow C_6H_6 + (CH_3)_3COMgBr$$

88. Answer (3)

$$\underset{CH_3}{\overset{CH_3}{>}}\underset{\substack{|\\ OH}}{C}-\underset{\substack{|\\ OH}}{CH}-C_2H_5 \xrightarrow{HIO_4} \underset{CH_3}{\overset{CH_3}{>}}C=O + H-\overset{\substack{O\\ ||}}{C}-C_2H_5$$

Oxidative cleavage of cis-vicinal diol is done by HIO_4.

89. Answer (1)

For isomeric alkyl halides

$$\text{B.P.} \propto \text{surface area} \propto \frac{1}{\text{number of branches}}$$

90. Answer (2)

In Swarts reaction, C–F bond formation takes place.

91. Answer (4)

$$CH_3-\overset{\substack{O\\ ||}}{C}-OC_2H_5 + CH_3MgBr \longrightarrow CH_3-\overset{\substack{O\\ ||}}{C}-CH_3 \xrightarrow{CH_3MgBr} CH_3-\overset{\substack{OH\\ |}}{\underset{\substack{|\\ CH_3}}{C}}-CH_3$$

3° - Alcohol

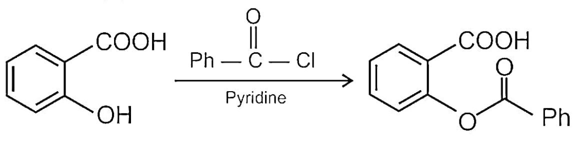
92. Answer (3)

Ethanal and ethanol cannot be distingushed by iodoform reaction as both of them gives yellow precipitate on iodoform reaction.

93. Answer (2)

Secondary alcohol gives blue colouration in Victor Meyer's test.

94. Answer (2)



Major

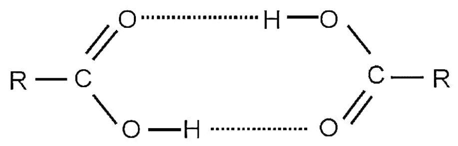
95. Answer (4)

$$\underset{(X)}{CH_3-\underset{\substack{|\\ OH}}{CH}-CH_3} \xrightarrow[H^+]{K_2Cr_2O_7} \underset{(Y)}{CH_3-\underset{\substack{||\\ O}}{C}-CH_3}$$

Y gives iodoform test with I_2/NaOH

96. Answer (1)

Most carboxylic acids exist as dimer in the vapour phase or in the aprotic solvents.

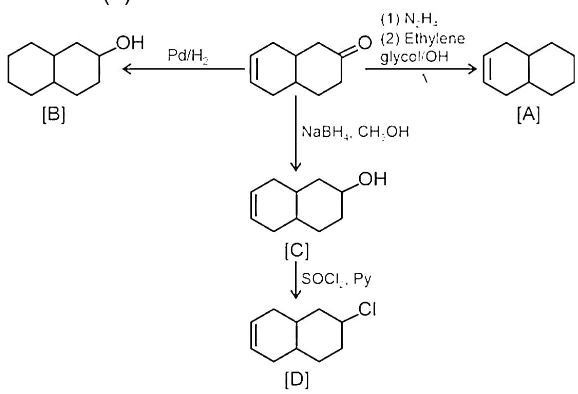


dimer
In vapour state or in aprotic solvent

97. Answer (2)

Due to ortho effect of $-CH_3$ group in benzoic acid its acidity increases.

98. Answer (4)



99. Answer (3)

Phenol gives violet colour with neutral $FeCl_3$.

100. Answer (2)

3° alcohol dehydrates on heating with Cu at 573 K.

BOTANY

SECTION - A

101. Answer (4)

Verhulst-Pearl logistic growth is described by the equation $\frac{dN}{dt} = rN\left(\frac{K-N}{K}\right)$.

102. Answer (2)

Some snails and fishes are known to enter into aestivation, a stage of suspended development.

103. Answer (2)

This graph is indicating response shown by conformers. Majority of animals and nearly all plants are conformers. They cannot maintain a constant internal environment.

104. Answer (4)

Lichens represent mutualistic relationship between a fungus and photosynthesising alga.

105. Answer (1)

A population has certain attributes whereas individual organism does not. An individual may have births and deaths but population has birth rates and death rates.

106. Answer (2)

Yellow-billed oxpecker sit on the back of black rhinoceros and feed on the ticks. This shows protocooperation type of interaction. Rest all are examples of commensalism.

107. Answer (4)

Birth rate = $\frac{\Delta N}{N\Delta t}$

$= \frac{15}{30} = 0.5$

Therefore, 0.5 offspring are added per year

108. Answer (3)

Resources need not to be limiting for competition to occur as feeding efficiency of one species might be reduced due to inhibitory presence of the other species. This is called interference competition.

109. Answer (1)

Parasites shows high reproductive capacity.

110. Answer (3)

Predators are ‘prudent’ in nature.

111. Answer (2)

Equation $\frac{dN}{dt} = rN$ describes exponential growth resulting in a J-shaped curve.

whereas, equation $\frac{dN}{dt} = rN\left(\frac{K-N}{K}\right)$ describes logistic growth resulting in a sigmoid curve.

112. Answer (1)

In the given question

Number of births (B) = 200

Number of deaths (D) = 180

Immigration (I) = 20

Emigration (E) = 30

Therefore, B + I = 200 + 20 = 220

D + E = 180 + 30 = 210

Since (B + I) > (D + E), the population size is increasing.

113. Answer (3)

Mean annual precipitation in grasslands is about 25 cm to 100 cm.

114. Answer (1)

Temperature, light, water and soil are the most important key elements that lead to so much variations in the physical and chemical conditions of different habitats.

115. Answer (2)

Each species has a distinct niche and no two different species occupy exactly the same niche.

116. Answer (4)

Salt concentration of sea water is 30-35 ppt.

117. Answer (1)

Eurythermal organisms can tolerate wide range of temperatures. Most of the mammals and birds are eurythermal.

118. Answer (3)

At higher altitudes, the binding capacity of haemoglobin with oxygen is decreased.

119. Answer (4)

Cuckoo bird laying its eggs in crow's nest shows brood parasitism.

120. Answer (3)

An assemblage of population of different species forms biological community.

Ecosystem is composed of biological community.

121. Answer (4)

In anaerobic sludge digester, anaerobic methanogenic bacteria produce a mixture of gases such as CH_4, H_2S and CO_2. These gases collectively form biogas.

122. Answer (2)

The Ministry of Environment and Forest has initiated Ganga and Yamuna action plan to save major rivers of our country from pollution.

123. Answer (1)

Streptokinase is produced by the bacterium *Streptococcus* and is modified by genetic engineering and used as 'Clot buster' for removing clots from the blood vessels of patients who have undergone myocardial infarction.

124. Answer (4)

Conversion of milk to curd improves its nutritional value by increasing the amount of vitamin B_{12}.

125. Answer (2)

During primary treatment in sewage treatment plants, the solids settle to form the primary sludge and the supernatant form primary effluent.

126. Answer (3)

Biogas can be used as source of energy as it is inflammable.

127. Answer (2)

Aerobic sludge contain aerobic heterotrophic microbes.

128. Answer (1)

Bacillus thuringiensis (Bt) is a microbial biocontrol agent that can be introduced to control butterfly caterpillars.

Ladybird is useful in controlling aphids.

Majority of baculoviruses used as biocontrol agent are in the genus *Nucleopolyhedrovirus*.

Trichoderma species are free living fungi that are common in root ecosystems.

129. Answer (3)

Cyanobacteria such as *Anabaena*, *Nostoc* and *Oscillatoria* are nitrogen fixing photosynthetic organisms. They add organic matter to the soil, increase its fertility and reduce alkalinity of soil.

130. Answer (3)

Plants having mycorrhizal association show resistance to root borne pathogens, tolerance to salinity and drought; and overall increase in plant growth and development. The fungal symbiont absorbs phosphorus from soil and passes it to plants.

131. Answer (3)

Wine and beer are naturally fermented and are not produced by distillation.

132. Answer (2)

The source for Penicillin is *Penicillium notatum*.

133. Answer (3)

This is a method of controlling pests that relies on natural predation. Chemical fertilisers are toxic and harmful to humans as they have been polluting our environment.

134. Answer (1)

Toddy is traditional drink of south India. It is made by fermenting sap from palm.

135. Answer (4)

Keoladeo national park is situated at Bharatpur and is famous for hosting Siberian cranes in winter.

SECTION - B

136. Answer (2)

Dosa and Idli are fermented preparation of rice and black gram, prepared by using bacteria.

137. Answer (1)

Antibiotic means against life, in context of disease causing organisms; whereas with reference to human beings, they are 'pro-life' and not against.

138. Answer (1)

Statins resemble mevalonate and is competitive inhibitor of HMG CoA reductase. It is commercialised as blood-cholesterol lowering agent. These are produced by yeast *Monascus purpureus*.

139. Answer (2)

Butyric acid is obtained from a bacterium and used for making rancid butter.

140. Answer (3)

The biological process in sewage treatment involves:

- Treatment of primary effluent with aerobic bacteria
- Sedimentation of flocs

141. Answer (1)

Flocs are masses of aerobic bacteria associated with fungal filaments to form mesh like structures.

142. Answer (2)

The greater the BOD of waste water, more is its polluting potential.

143. Answer (1)

In logistic growth, a population growing in a habitat with a limited resources shows initially a lag phase followed by phase of acceleration and deceleration and finally an asymptote.

144. Answer (3)

Kangaroo rat in North American deserts is capable of meeting all of its water requirement through its internal fat oxidation where water is released as by product.

145. Answer (4)

Mortality and emigration will decrease the population density.

146. Answer (2)

The 'r' in the $\frac{dN}{dt} = rN$ equation is called the 'Intrinsic rate of natural increase' and is a very important parameter chosen for assessing impacts of any biotic or abiotic factors on population growth.

147. Answer (3)

Graphical representation of age structure is a characteristic of population.

148. Answer (2)

Competition is a (–, –) interaction.

149. Answer (4)

Exponential growth is represented by equation $\frac{dN}{dt} = rN$, resulting in J-shaped growth curve. Logistic growth model is more realistic one.

150. Answer (2)

Mammals from colder climates generally have shorter ears and limbs to minimise heat loss. In the polar sea, a thick layer of fat is found in aquatic mammals like seals below their skin.

ZOOLOGY

SECTION - A

151. Answer (3)

Transmission of HIV infection generally occurs :-

(a) By Sexual contact with an infected person

(b) By transfusion of contaminated blood and blood products

(c) By sharing infected needles

(d) From infected mother to her foetus through placenta

152. Answer (4)

Radiotherapy is used in the treatment of cancer alongwith surgery and chemotherapy.

Radiography, CT and MRI are techniques that are used in detection of cancer of the internal organs.

153. Answer (3)

Aedes mosquitoes are vectors of pathogens of diseases like dengue and chikungunya.

Anopheles is the vector of *Plasmodium* while *Culex* is the vector of *Wuchereria*.

154. Answer (1)

Microsporum is a genus of fungi. Heat and moisture help them to grow and thrive in groin or toes. These fungi are responsible for ringworms. Ringworms are generally acquired from soil or by using towels, clothes, comb, *etc.* of infected individuals.

155. Answer (2)

A protozoan, *Plasmodium falciparum* is responsible for causing malignant malaria.

156. Answer (3)

Rhino viruses cause common cold. They infect the nose and respiratory passage but not the lungs.

157. Answer (2)

The discovery of blood circulation by William Harvey using experimental method and the demonstration of normal body temperature in persons with black-bile using thermometer disproved the 'good humor' hypothesis of health.

158. Answer (3)

Balanced diet, personal hygiene and regular exercise are very important to maintain good health. Yoga has been practiced since time immemorial to achieve physical and mental health. A sedentary lifestyle has an array of adverse health effects.

159. Answer (3)

Incidences of vector-borne (*Aedes* mosquito) diseases like dengue and chikungunya are widespread in many parts of India.

160. Answer (1)

Formation of gametocytes occur in human RBCs which are picked up with blood meal by female *Anopheles* mosquito.

161. Answer (3)

The primary lymphoid organs are bone marrow and thymus.

162. Answer (4)

A wide range of organisms (not all), belonging to bacteria, viruses, fungi, protozoan, helminths, *etc.* can cause diseases in man and are called pathogens. For example, *Paramoecium*, *Entamoeba coli* and many species of yeast are non-pathogenic.

163. Answer (2)

Diseases which are easily transmitted from one person to another, are called infectious diseases. Cancer is a non-infectious disease. Infectious diseases like AIDS are fatal.

164. Answer (3)

Salmonella typhi is a pathogenic bacterium (moneran) which causes typhoid fever in human beings. These pathogens generally enter the small intestine through contaminated food and water. Colon is the part of large intestine. Typhoid fever can be confirmed by Widal test.

165. Answer (4)

Antibody binding site is the part of antigens. Antibody contains antigen binding site.

166. Answer (2)

IgA is found abundantly in colostrum. IgG crosses placenta during pregnancy.

167. Answer (3)

Opiate narcotics – Opium, morphine, heroin, pethidine, methadone *etc.* Hashish and marijuana are cannabinoids.

168. Answer (1)

The lining of the major tracts (respiratory, digestive and urogenital tracts) have lymphoid tissue termed as MALT (Mucosa Associated Lymphoid Tissue).

169. Answer (1)

The malarial parasites initially multiply within the liver cells and then attack the RBCs resulting in their rupture. The rupture of RBCs is associated with the release of a toxic substance, haemozoin, which is responsible for the chill and high fever recurring every three to four days.

170. Answer (4)

Haemophilus influenzae is a bacterium which causes pneumonia. *Trichophyton* is a fungal genera that causes ringworms.

Common cold is caused by Rhino virus. It is characterised by nasal congestion and discharge, sore throat, cough, headache, *etc*.

171. Answer (2)

The spleen is a large bean-shaped organ. It mainly contains lymphocytes and phagocytes. It acts as a filter of the blood by trapping blood-borne microorganisms.

172. Answer (2)

AIDS is an STI and is not curable. Bronchitis is inflammation of bronchi.

173. Answer (3)

In a monomeric antibody molecule, the total number of intrachain disulphide bonds on both heavy chains is 8 and on both light chains is 4.

174. Answer (4)

Given figure represents poppy plant. Morphine is extracted from the latex of this plant. Morphine belongs to opioids. It is generally taken by snorting and injection. Cannabinoids are generally taken by inhalation and oral ingestion. Coca plant is native to South America.

175. Answer (4)

Heroin, commonly called smack is a white, odourless, bitter crystalline compound, whose receptors are present in our CNS and GIT. It is generally taken by snorting and injection. Heroin is a depressant and slows down the body functions.

176. Answer (2)

The cancer patients are treated with substances called biological response modifiers such as α-interferons which activates their immune system and helps in destroying the tumor.

177. Answer (3)

The symptoms of pneumonia include fever, chills, cough and headache. In severe cases, the lips and finger nails may turn gray to bluish in colour.

178. Answer (3)

Adolescence is a bridge linking childhood and adulthood.

179. Answer (3)

First generation vaccines include attenuated live vaccines and killed/inactivated vaccines. Second generation vaccines are made using surface antigens of the pathogens *e.g.*, Hepatitis B vaccine. Synthetic vaccines are third generation vaccines.

180. Answer (3)

Smoking reduces the concentration of haembound oxygen. This causes oxygen deficiency in the body.

181. Answer (2)

Sporozoites are stage of *Plasmodium* that enters the human body through infected female mosquito bite. Gametocytes are produced in RBCs of an infected human. Female mosquito takes up these gametocytes with blood meal.

182. Answer (4)

The spleen is a large bean-shaped organ. It acts as a filter of the blood by trapping the blood-borne micro-organisms. Thymus and bone-marrow are primary lymphoid organs.

183. Answer (2)

Innate immunity consists of four types of barriers. These are physical, physiological, cellular and cytokine barriers. Mucus coating of the epithelium lining the respiratory, gastrointestinal and urogenital tracts are physical barriers of innate immunity.

184. Answer (2)

Acquired immunity is characterised by memory. This means when our body encounters a pathogen for the first time it produces primary response which is of low intensity. Subsequent encounter with same pathogen elicits a highly intensified secondary or anamnestic response.

185. Answer (4)

Antibodies are secreted by B-lymphocytes.

Monocytes and neutrophils are phagocytic cells.

SECTION - B

186. Answer (3)

Education and counselling help youngsters to face problems and stresses, and to accept disappointments and failure as a part of life.

187. Answer (3)

Filariasis/elephantiasis is caused by *W. bancrofti* and *W. malayi*. It is a slow developing chronic inflammation of the organs in which the pathogens live for many years, usually the lymphatic vessels of the lower limbs. The genital organs are also often affected.

188. Answer (2)

Ascaris is an intestinal parasite causes ascariasis. Symptoms of this disease include internal bleeding, muscular pain, fever, anemia and blockage of the intestinal passage.

189. Answer (3)

Lack of interest in personal hygiene and deteriorating relationship with family and friends are the common warning signs of drug abuse among youth.

190. Answer (1)

Lysozyme that is present in sweat, saliva and tears, destroys certain types of bacteria.

191. Answer (4)

Virus infected cells secrete proteins called interferons which protect non-infected cells from further viral infections.

Interferons are cytokine barriers of innate immunity.

192. Answer (2)

Symptoms mentioned are of amoebiasis. *Entamoeba histolytica* is a protozoan parasite in the large intestine of human.

193. Answer (3)

Cannabinoids are known for their effects on cardiovascular systems of the body.

194. Answer (2)

Filariasis is transmitted through the bite by the infected female mosquito vectors to a healthy person.

195. Answer (3)

Allergy is caused due to release of chemicals like histamine, serotonin from mast cells. Adrenaline is secreted by adrenal medulla, which quickly reduces the symptoms of allergy.

196. Answer (3)

T-lymphocytes are involved in cell mediated immunity, while B-lymphocytes are involved in humoral immunity. Platelets help in blood clotting and RBCs help in the transport of gases.

197. Answer (3)

Drugs like barbiturates, amphetamines, benzodiazepines are normally used as medicines to help patients cope with mental illness like depression and insomnia.

198. Answer (2)

Tissue matching, blood group matching are essential before undertaking any graft. The body is able to differentiate self and non-self and the cell mediated immune response is responsible for the graft rejection.

199. Answer (3)

Addiction is a psychological attachment to certain effects–such as euphoria–associated with drugs and alcohol. With repeated use of drugs, the tolerance level of the receptors present in our body increases. Dependence leads the patient to ignore all social norms in order to get sufficient funds to satiate his/her needs.

200. Answer (3)

The side-effects of the use of anabolic steroids in human females include masculinisation, mood swings, depression, enlargement of clitoris, *etc.* Breast enlargement is one of the side effects of the use of anabolic steroids in human males.

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